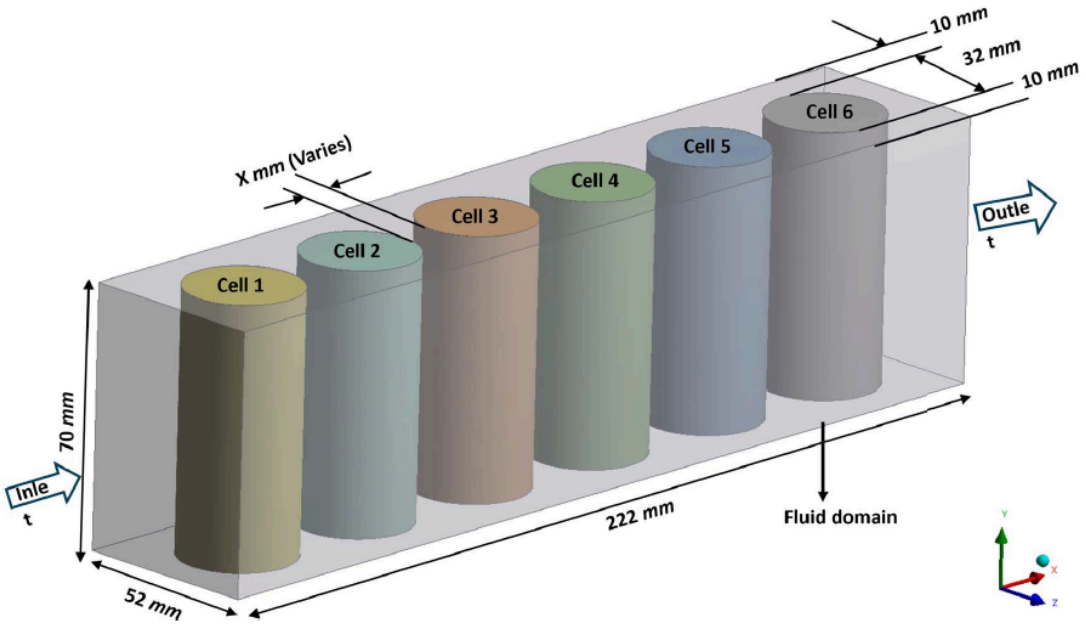
Battery Thermal Management System Based on Phase Change Material and Air Convection Cooling.

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Introduction:

Presently Electric vehicles are having more advantage than Internal combustion engine vehicles due to zero emissions, robust performance, low noise pollution and less cost for maintenance. **Battery** is one major component in EVs which decides the EVs overall performance, a reversible electrochemical battery has cyclic charging and discharging capabilities which provides a portable application for EV batteries. For both small- and large-scale applications incorporation, many technical improvements were made with respect to high specific energy density, nominal voltage, high life cycle, and low self-discharge. **Lithium-ion (Li-ion) batteries** are widely used in light and heavy-duty vehicles in present days.



Li-ion batteries are well known for their heat generation characteristics and resulting rise in the battery temperature especially during high charge and high discharge rates. The main reason for the degradation of battery performance at low temperatures is the **change in internal properties** of the electrolyte

Objective:

To analyze the thermal behavior of an EV battery cell integrated with **PCM (OM35)** and casing under **convective cooling conditions**, with the aim of evaluating **temperature distribution**, **heat dissipation** effectiveness, and the role of PCM in maintaining **safe operating temperatures.**

Methodology:

The geometry of the EV battery cell module was modeled as a composite structure consisting of three **concentric cylindrical layers** to accurately represent the physical configuration of the system. The innermost cylinder corresponds to the battery cell, having a diameter of **32 mm**. Surrounding the cell is a layer of phase change material (PCM) with a uniform thickness of **6 mm,** which is incorporated to enhance thermal regulation by absorbing excess heat during operation. The outermost layer represents the protective casing, modeled with a thickness of **2 mm,** providing structural integrity and facilitating heat dissipation to the surrounding environment.

This concentric arrangement ensures a realistic representation of the thermal interaction between the cell, PCM, and casing, enabling effective analysis of heat transfer characteristics within the system.

OM35 is a paraffin-based phase change material with a melting point around **35 °C**. It absorbs **excess heat** through latent heat during phase change, helping to control temperature rise in battery systems. It is thermally stable, chemically inert, and suitable for repeated thermal cycles.

The properties of a battery cell & PCM material are given below.

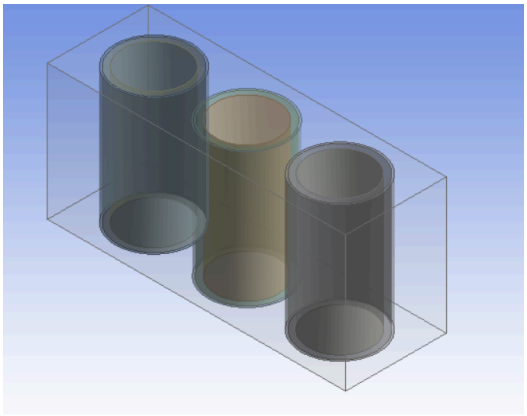
Table 1 Electrical and thermophysical properties of the battery cell

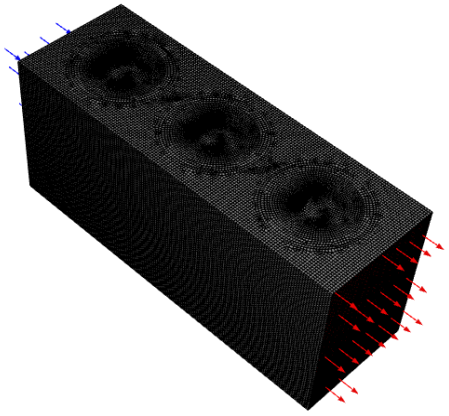
Parameters	Details
Cell diameter, mm	32 ± 0.3
Cell length, mm	70 ± 0.2
Capacity, Ah	6
Nominal voltage, V	3.6
Cell mass, g	145
Mass density, kg/m^3	2879
Thermal conductivity, W/m-K [61]	0.743
Specific heat, J/kg-K	880

Table 2 Thermophysical and electrical properties of PCM material

PCM (OM35)	
Parameters	Details
Solid density, kg/m^3	900
Liquid density, kg/m^3	870
Specific heat, J/kg-K	2570
Solid thermal conductivity, W/m-K	0.2
Liquid thermal conductivity, W/m-K	0.16
Viscosity, kg/m-s	0.001372
Pure solvent melting heat, J/kg	171,000
Solidus temperature, °C	33
Liquid temperature, °C	34
Basic material	Organic
Thermal stability, cycle	~2000
Maximum operating temperature, °C	120

The geometry was developed in ANSYS Workbench, and the meshing was carried out using ANSYS Fluent.





These are the building geometry of the EV Battery Cell

Parameters & Initialization:

Solver Type: Transient

- **Description:** The simulation is time-dependent, capturing changes in flow and temperature with respect to time.
- **Why used:** Battery thermal behavior (heating, cooling, PCM melting/solidification) is inherently **unsteady**, especially during charge/discharge cycles. Steady-state would miss these dynamics.

Solver: Pressure-Based

- **Description:** Solves flow using pressure–velocity coupling.
- **Why used:** Suitable for **low-speed flows (air cooling)** in battery systems where density variations are small. It ensures stable and accurate solutions for incompressible flow.

Energy Equation: Enabled

- **Description:** Governs heat transfer within the domain.
- **Why used:** Essential for predicting **temperature distribution** in the battery, casing, air, and PCM. Without this, thermal analysis is not possible.

Viscous Model: Enabled

- **Description:** Accounts for fluid viscosity and turbulence effects (depending on model selected).
- **Why used:** Airflow inside battery packs is often **laminar or weakly turbulent**, and viscosity influences heat transfer and velocity distribution.

Solidification/Melting Model: Enabled

- **Description:** Models phase change between solid and liquid states.
- **Why used:** Critical for **PCM (Phase Change Material)** analysis, as it absorbs heat during melting and releases during solidification, improving thermal management.

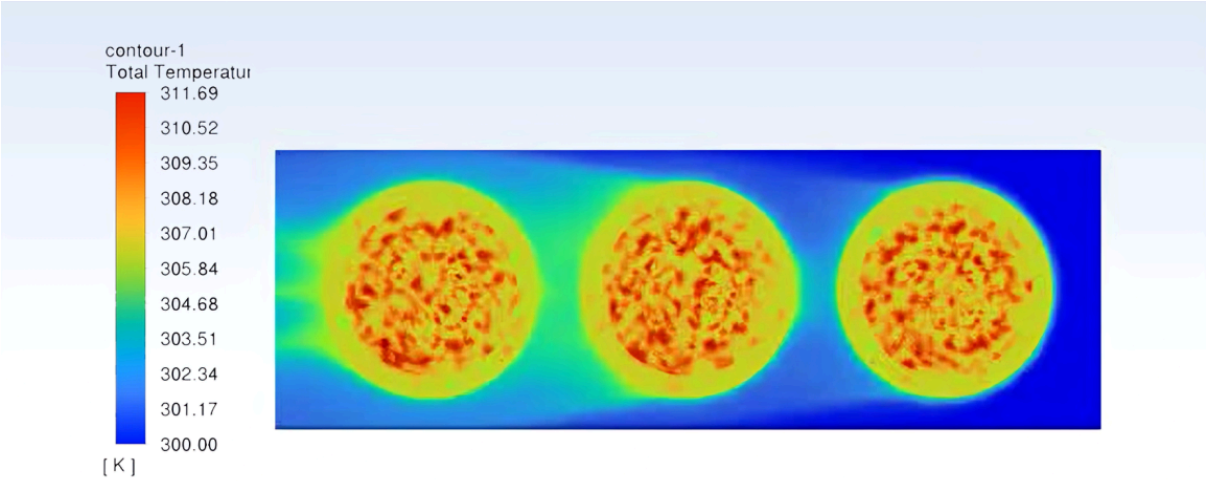
Cell	Fluid	Li-battery
PCM	Fluid & solid	PCM-Solid & PCM-Liquid

Casing	solid	copper

Results (Materials):

Number of iterations required to achieve convergence.

Temperature contour:



Observations:

Maximum temperature (~311–312 K) is observed at the **cell cores**, indicating significant internal heat generation.

A strong **radial temperature gradient** is present (cell → PCM → casing → air), showing effective heat conduction outward.

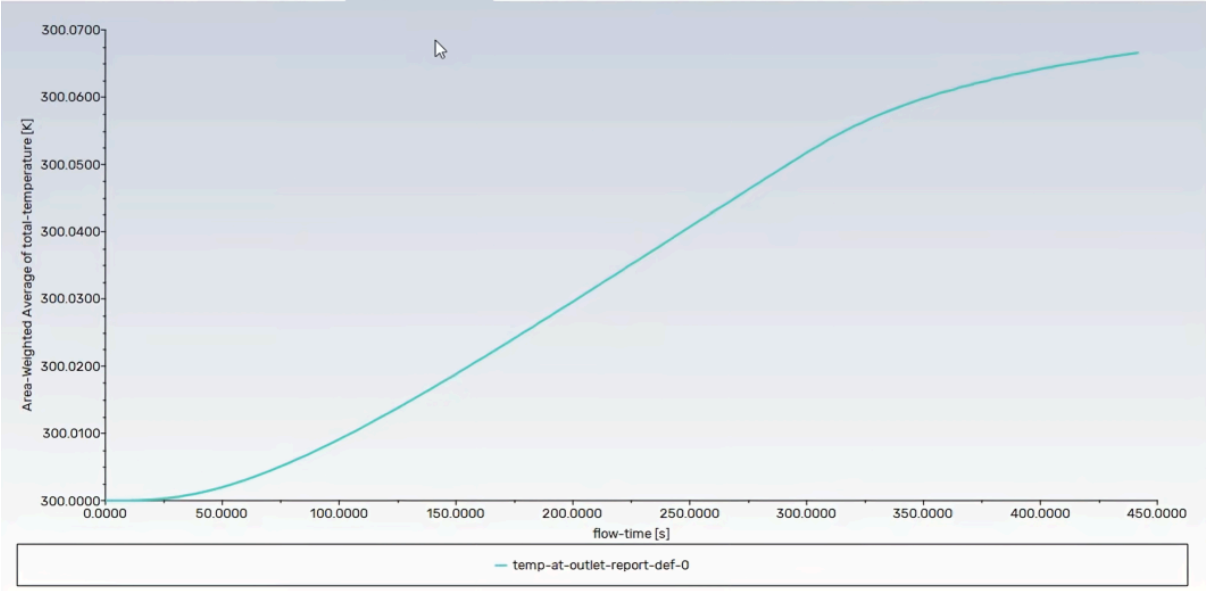
The PCM region maintains **moderate temperatures (~304–307 K)**, indicating active heat absorption and thermal buffering.

The air region shows **clear cooling (300–303 K)**, confirming effective convective heat removal.

Slight temperature non-uniformity between cells suggests **interaction effects and uneven heat dissipation**.

Overall higher temperature range compared to earlier case indicates **increased thermal load or progressed transient heating**.

Temperature at Outlet



Observations:

- Outlet temperature shows a gradual increase with time, indicating **continuous heat accumulation** in the system.
- The curve is initially slow rising, suggesting thermal inertia and **delayed heat propagation.**
- A near-linear increase region (mid-time) indicates steady heat transfer from cell to surroundings.
- The slope reduces at later times, showing a tendency toward thermal stabilization.
- Overall temperature rise is small **(~0.07 K),** indicating effective cooling and controlled heat removal.

Conclusions:

- The temperature contours show that heat is generated at the **cell core** and effectively dissipated outward, confirming proper heat transfer through PCM and casing.
- PCM acts as a **thermal buffer**, reducing peak temperatures and maintaining a more uniform temperature distribution.
- The outlet temperature rise is gradual and limited, indicating that the cooling system **effectively removes heat.**
- The system shows a trend toward thermal stability, with **no extreme hotspots** observed.
- Overall, the combined PCM and air cooling setup provides **efficient thermal management** under given operating conditions.

References:

1. Swamy, Kundrapu Ayyappa, and Saket Verma. "Experimental and numerical investigation for optimization of a hybrid battery thermal management system based on phase change material and air convection." *Journal of Thermal Science and Engineering Applications* 16.12 (2024).

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